

Swathi Abhigna Vemula

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EDUCATION

SRM University AP, Andhra Pradesh

2023 – 2027

Bachelor of Technology in Computer Science and Engineering, specialization in AI & ML

CGPA: 8.98

EXPERIENCE

SRM University AP

May 2025 – Aug 2025

Research Intern — Generative AI & Workflow Automation

Amaravati, AP

- Assisted in the design and implementation of AI-powered workflow automation pipelines for legal operations, compliance management, and document processing using Python orchestration frameworks
- Reduced manual document processing effort by automating end-to-end review and classification pipelines, improving operational efficiency across legal and compliance workflows
- Contributed to faster delivery timelines across multiple simultaneous tracks by standardizing documentation and prototype testing processes

PROJECTS

Edge-Enabled Deepfake and Image Forgery Detection System | *CNN, Computer Vision*

- Developed a lightweight CNN-based pipeline for real-time deepfake and image forgery detection, engineered for edge deployment under strict compute constraints with no cloud dependency

AI Resume Screening & Candidate Ranking System | *Machine Learning, NLP*

- Built an NLP-based resume screening and ranking system using TF-IDF vectorization, Logistic Regression, and cosine similarity scoring to match candidates against job descriptions

Movie Script Rating Prediction | *NLP, Regression*

- Developed an NLP-based regression model using TF-IDF and Linear SVR to predict movie ratings directly from raw screenplay text

PARTICIPATIONS & ACHIEVEMENTS

Smart India Hackathon 2025 — Finalist | *Hardware Edition*

- Reached the national finals as part of team TEDH; built SandSight — an autonomous three-wheeled rover for beach sand grain-size mapping using a camera-based imaging setup with controlled lighting, and CV/ML pipelines to estimate key grain-size metrics (D10, D50, D90)
- Delivered a geospatial platform with backend inference APIs, an interactive real-time dashboard, and Firebase-based authentication, cloud storage, and database sync — integrating hardware mobility, ML inference, and frontend visualization under tight hackathon timelines

SKILLS

Languages : C, Python, JavaScript, SQL

Frameworks & Frontend : HTML, CSS, React.js

AI/ML : Machine Learning, Deep Learning, Generative AI, NLP, Computer Vision, LLMs, Prompt Engineering

Tools & Platforms : OpenCV, GitHub, REST APIs, Jupyter Notebook

Methodologies : Data Pipeline Design, Model Evaluation, Technical Documentation